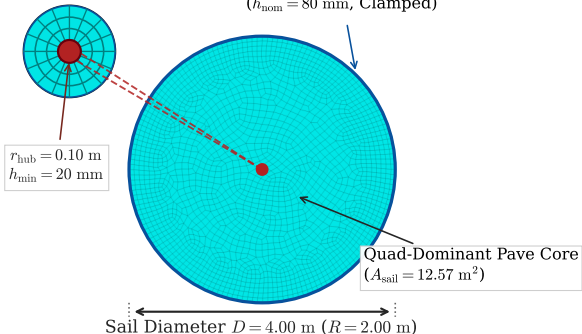


Detail A: Payload Hub Interface

Structured Perimeter Band
($h_{\text{nom}} = 80 \text{ mm}$, Clamped)



ANSYS FEA Discretization Metrics (SHELL181):

- Element Type: 4-Node Structural Shell (Mindlin-Reissner, Large Deflection)
 - Mesh Density: 3,048 Elements || 3,112 Nodes
 - Sizing: $h_{\text{nom}} = 80 \text{ mm}$ (Rim) $\rightarrow$ $h_{\text{min}} = 20 \text{ mm}$ (Payload Hub Interface)
- Mesh Quality: Min Orthogonal Quality = 0.942, Average Aspect Ratio = 1.12